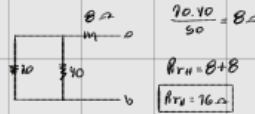


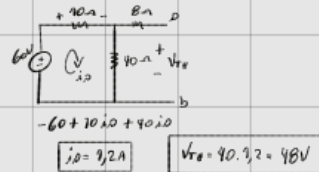
Divisor de tensão: $V_x = V_{total} \cdot \left(\frac{R_x}{R_{eq}} \right)$

Divisor de corrente: $I_x = I_{total} \cdot \left(\frac{R_{eq}}{R_x} \right)$
 ↳ do que queramos.

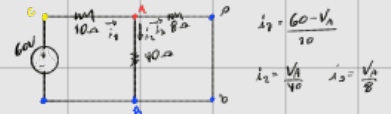
Resistência de Thevenin:



* Tensão de TH



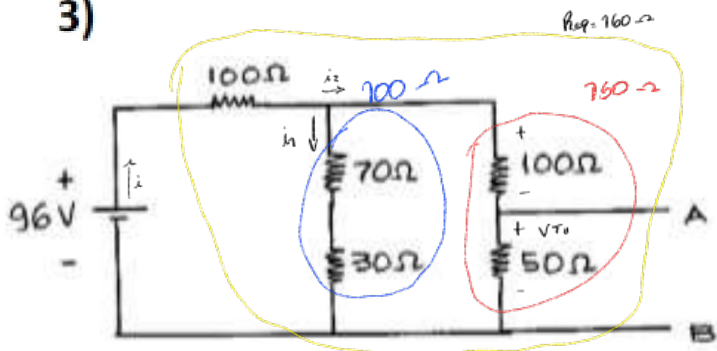
Corrente de TH



Nó A: $i_1 = i_2 + i_3 \Rightarrow \frac{60 - V_A}{10} = \frac{V_A}{40} + \frac{V_A}{8} \times 90$
 $240 - 4V_A = V_A + 5V_A \Rightarrow V_A = 24V$

$i_3 = I_{TH} = \frac{24}{8} = 3A \Rightarrow I_{TH} = 3A$

3)



Cálculo de tensão de TH:

$96 = 160 \cdot i$

$i = 0,6A$

$i_1 = 96 \cdot \frac{60}{100}$

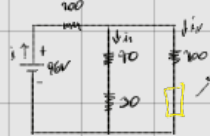
$i_1 = 0,36A$

$i_2 = 0,24A$

$V_{TH} = 0,24 \cdot 50$

$V_{TH} = 12V$

Cálculo de I_{NO}



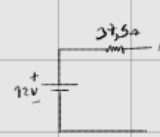
$96 = 150 \cdot i \Rightarrow i = 0,64A$

Tensiones, pois
 ligamos o
 curto-circuito.

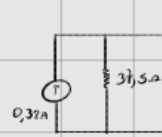
$I_{NO} = 0,32A$

Equivalências:

THEVENIN:



Norton:

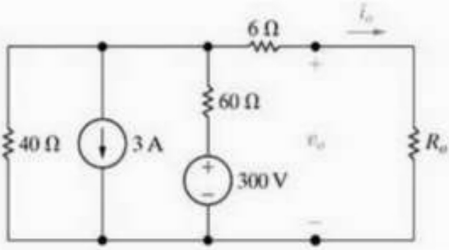


$V_{TH} = R_{TH} \cdot I_N$

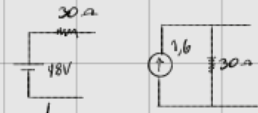
$R_{TH} = \frac{V_{TH}}{I_N}$

$R_{TH} = \frac{12}{0,32} = 37,5\Omega$

4) Para o circuito a seguir, determinar i_0 e v_0 , considerando os seguintes valores para R_0 : 2 Ω, 10 Ω, 50 Ω e 120 Ω.

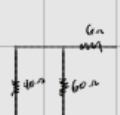


* Calcular um circuito equivalente
 para facilitar no cálculo com os
 resistores diferentes



↳ Logo, use o cálculo a que se pede no questionário.

Cálculo de R_{TH} :



Cálculo de V_{TH} :



Análise nodal:

$V_0 = 300V$

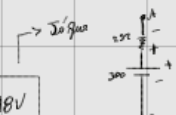
* Nó A

$\frac{V_A}{40} + \frac{V_A - V_0}{60} + 3 = 0$

$V_A = 98V$

$V_{TH} = 48V$

* Resistor de 6Ω está em aberto.



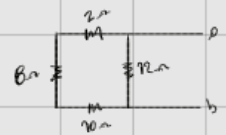
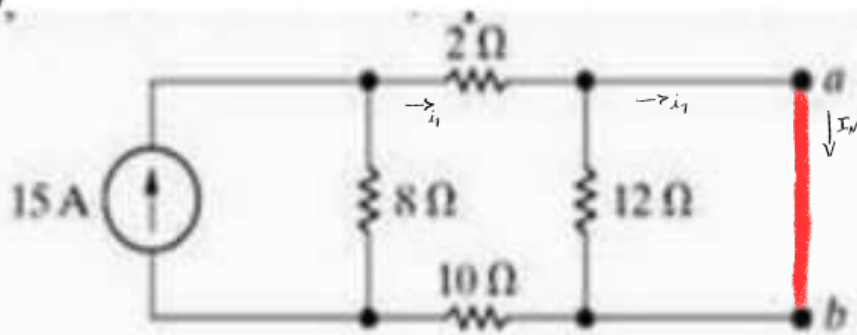
* Corrente de Norton:

$V_{TH} = R_{TH} \cdot I_N$

$I_N = \frac{V_{TH}}{R_{TH}} = 1,6A$

↳ Poderão fechar o curto circuito
 e criar um nó 0 mais para facilitar.

2)



$$R_{TH} = \frac{20 \cdot 12}{32} = 7,5 \Omega$$

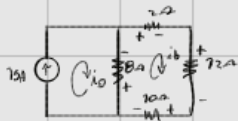
$$* i_x = i_1 \cdot \left(\frac{R_T}{R_x} \right)$$

$$i_1 = 15 \cdot \left(\frac{9,8}{12} \right) = 6A$$

$$I_N = 6A$$

Corr. 100 para modo no de 12V

* Calcula Tensão:



$$8(i_1 - i_2) + 2i_1 = 0$$

$$32i_1 - 8 \cdot 15 = 0$$

$$i_1 = 3,75$$

$$V_{TH} = 12 \cdot 3,75$$

$$V_{TH} = 45V$$